2026

FIRST EDITION

FAS BOOKLET

IGEM IIT BOMBAY INITIATIVE

A field guide to fungal fatty acid synthase, written by the team that is about to edit it.

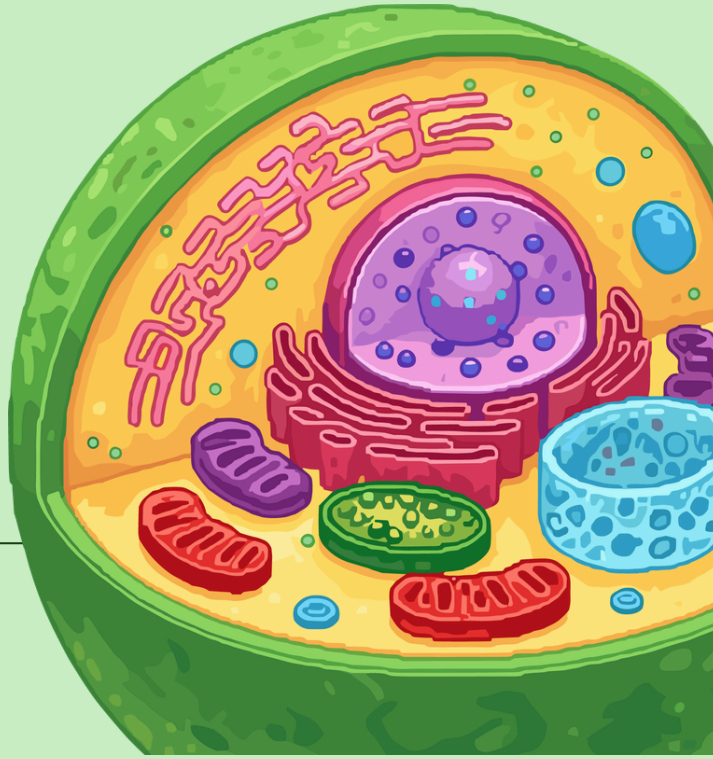


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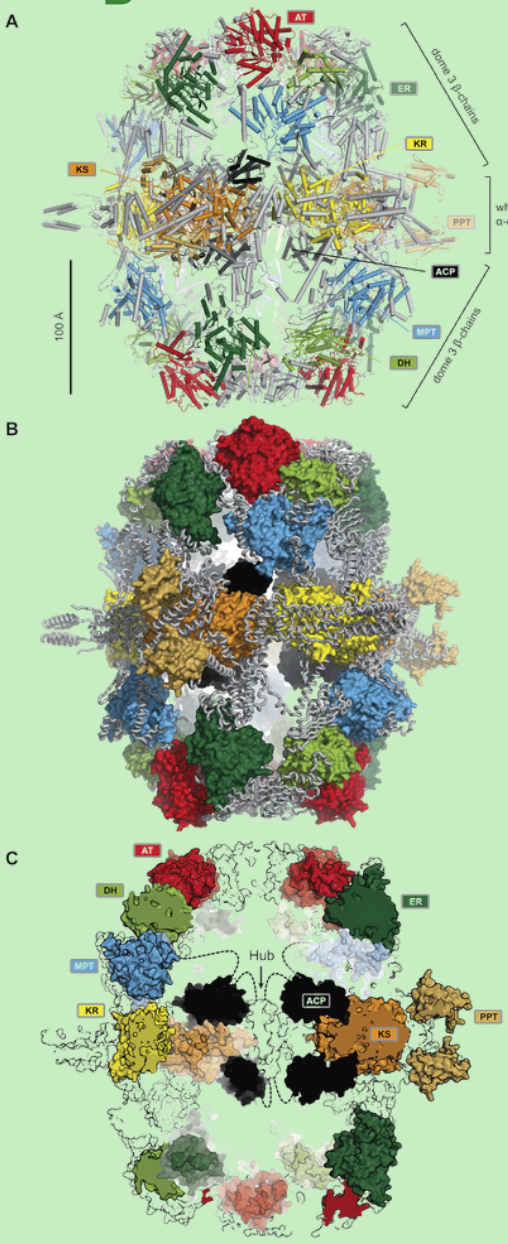
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Hi Reader, Welcome to the Molecular Factory!

Right now, in every one of your cells, a machine is assembling fat two carbons at a time. It started before you were born and it stops when you do.

The machine is fatty acid synthase (FAS), a 2.6-megadalton complex that builds a sixteen-carbon chain by repeating one four-step reaction cycle. Almost every organism has a version of it, and almost nobody has heard of it. This booklet dissects the fungal Type I FAS, the sealed α6β6 barrel of *Saccharomyces cerevisiae*.

We are iGEM IIT Bombay (Team BiFROST). This year we are editing a single residue of this machine so that yeast makes myristic acid (C14) instead of its usual palmitic (C16) and stearic (C18) acids. We wrote the booklet while learning the system, and we organised it around a method: each chapter views FAS through a different lens (a factory, a multicore chip, a four-stroke engine, a kinetic race, a control system, a tuning knob) and then pushes the lens until it breaks. Every break marks a physical principle the lens did not predict, and that is where the real biochemistry sits. Read critically, and argue with us.



Best regards,

Team BiFrost, iGEM IIT Bombay

Fatty Acid Synthase: Function and Cost

Every lineage that makes fat has had to solve the same problem: how do you build a long, greasy chain without losing it?

Fat synthesis is among the oldest jobs in metabolism. It runs in yeast, moulds, mushrooms, plants, bacteria and you, always by the same chemistry: build the chain two carbons at a time. What differs between lineages is the hardware.

In fungi the hardware is a barrel, 27 nm long and made of twelve protein chains, sealed. Inside it are an assembly line, an arm tethered to the wall, and a termination point that, in yeast, falls at sixteen carbons.

For our project we want fourteen. Fourteen carbons is myristic acid, and its methyl ester can serve as a data-centre coolant. Through this booklet we follow our own reasoning as we design the project, and we learn FAS along the way

Fatty Acids: Structure and Roles

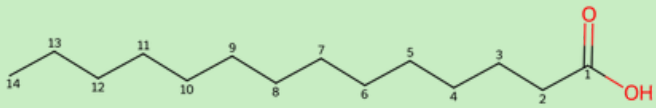
A fatty acid is a long aliphatic chain with a carboxylic acid group at one end. Because synthesis adds two carbons at a time, chains are almost always even-numbered; in most eukaryotic membranes and storage lipids they are 16 or 18 carbons. One end loves water, the other hates it, and that is why membranes exist. They are the cell's multi-tool: bricks, batteries and messengers.

Chain length and saturation set how tightly lipids pack, and so how fluid a membrane is, the same physics that makes butter solid and olive oil liquid. A cell that cannot control both cannot build a working membrane.

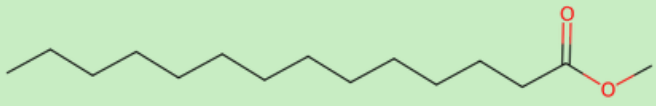
Fatty acids are expensive. One molecule of palmitic acid, the sixteen-carbon one, costs the cell the equivalent of 122 ATP (Singh et al., 2020). A process that costly cannot run open-loop, which is why Chapter 5 is about brakes.

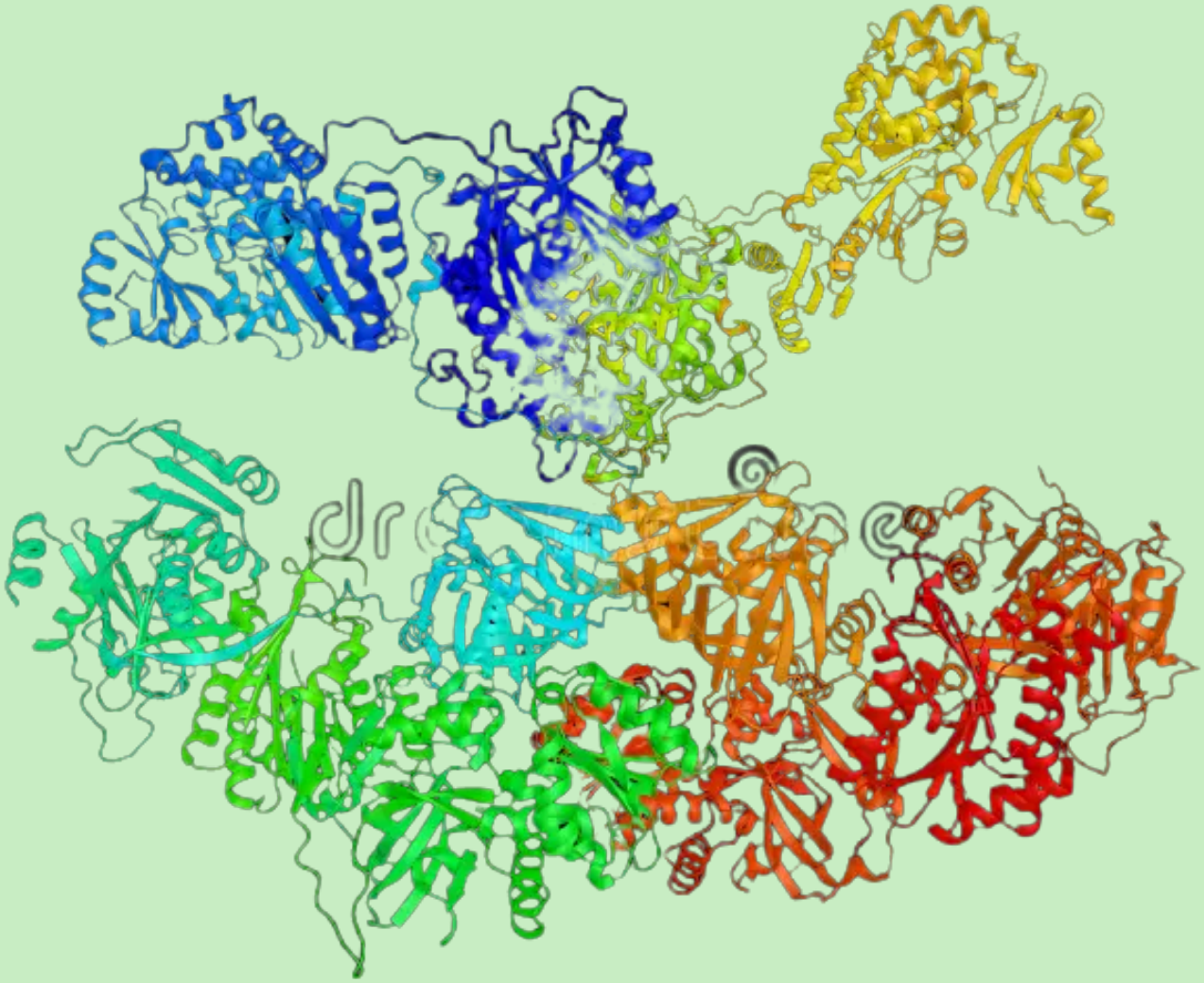
Reading map. Chapter 2 maps the hardware. Chapter 3 follows the chemistry of one cycle. Chapter 4 asks why the chain stops at sixteen. Chapter 5 covers the controls the cell already has. Chapter 6 is our design: what we are editing, why, and what we have built so far.

Myristic acid, C14:0 (14 carbons, fully saturated)



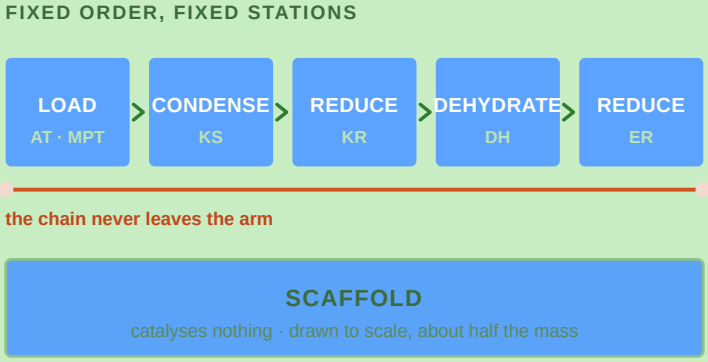
Methyl myristate (the same chain with a methyl ester on the acid end)





The Factory Model: A Useful Wrong Idea

We will call FAS a factory throughout this booklet, knowing it is wrong. Somewhere in each cell is an assembly line that never shuts down, never takes a shift off, and does not know you exist. It does one job, making fat, and it does it with more fidelity than any production line humans have built.



Work happens in a fixed order at fixed stations. FAS has eight functional domains (seven enzymes plus a carrier), and the chemistry forces them to run in sequence: load, condense, reduce, dehydrate, reduce.

There is a building. About half the mass of FAS is scaffold that catalyses nothing and exists to hold the working parts in position (Bukhari et al., 2014). It is the nearest thing a protein has to a factory floor plan.

And something physically carries the work from station to station: the acyl carrier protein (ACP) and its arm, met on page 12. That is where the model stops being kind. Turn the page.

Three Failures of the Factory Model

No conveyor: covalent tethering.
In a factory the product moves and the workers stand still. Here the product is bolted to the end of an arm by a single sulfur atom and stays attached through every step. Nothing is handed over; the arm swings and the enzymes lean in. An intermediate that is never released can never diffuse away and never needs to be found again. Chemists call this substrate channelling.

Failure 2-No manager: control by geometry and charge. Nobody directs the process. Each step happens because the arm reaches a station whose shape and charge suit the cargo. The decision procedure is the architecture, and there is no step above it. Biologists call this self-organisation.

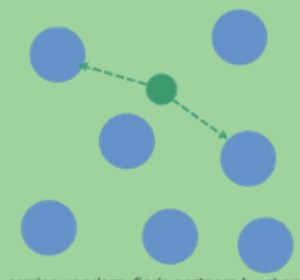
No calm: thermal noise is the engine. A factory is quiet; a protein in a warm cell is not. Water molecules strike it on the order of 10^{12} times a second from every direction. At this scale thermal energy (about 2.5 kJ/mol) is comparable to the energies of the interactions that hold the machine together, and viscosity dominates over inertia, so it is like swimming in honey. The arm is not driven to the next station. It is jostled, and the stations are positioned so that some of the jostling ends somewhere useful. Physicists call this a Brownian ratchet: noise is not a problem the machine works around, it is the power source.

The scale model. Blow the barrel up until it is as long as a football pitch and the arm becomes a 7-metre tether, the product an 8-metre rod, and the whole thing is shaken by an earthquake that never stops.

	FACTORY	vs	FAS
PRODUCT	Rides a belt		**Bolted to an arm, never released**
WORKERS	Stand still		**Stations stay put, the arm swings**
MANAGEMENT	Supervisors, schedules		**None: charge and geometry**
ENVIRONMENT	Calm		**Constant thermal bombardment**
BRAKES	Shift manager		CLOSEST MATCH **Three control systems (Chapter 5)** *the one place the analogy gets closer*

Bacteria and plants

7 separate proteins



carrier wanders, finds partners by chance

Fungi (our yeast)

1 sealed barrel, 12 chains



carrier is tethered inside, only swings

Animals (you)

1 chain, pairs into an X



two chains, open and looser

1. carrier searches a volume
2. carrier sweeps a surface
3. carrier on an open frame.

Type I vs Type II: Three Architectures for One Chemistry

Life has solved fatty acid synthesis three ways. Biochemists call the dissociated version Type II and the fused versions Type I. We compare them by one criterion: how the carrier protein finds its work. Our electronics metaphors are deliberately imperfect.

01

Bacteria and plants (Type II): the wireless network

Seven separate enzymes float free. A small carrier protein must collide with each one in turn, in the right order, by diffusion alone. That works poorly enough that bacteria keep a lot of carrier around just to improve the odds of a useful encounter.

02

Fungi (our yeast, Type I): the system-on-a-chip

Everything is hard-wired into one sealed barrel of twelve chains ($\alpha_6\beta_6$). The carrier is tethered inside and cannot leave, so it never searches; it only swings. Inside the barrel its effective concentration is around a millimolar, about ten times what bacteria manage

03

Animals (you, Type I): the modular board

Everything is fused onto one chain that pairs into an open X of about 540 kDa. It is looser than the barrel and tighter than the bacterial system.

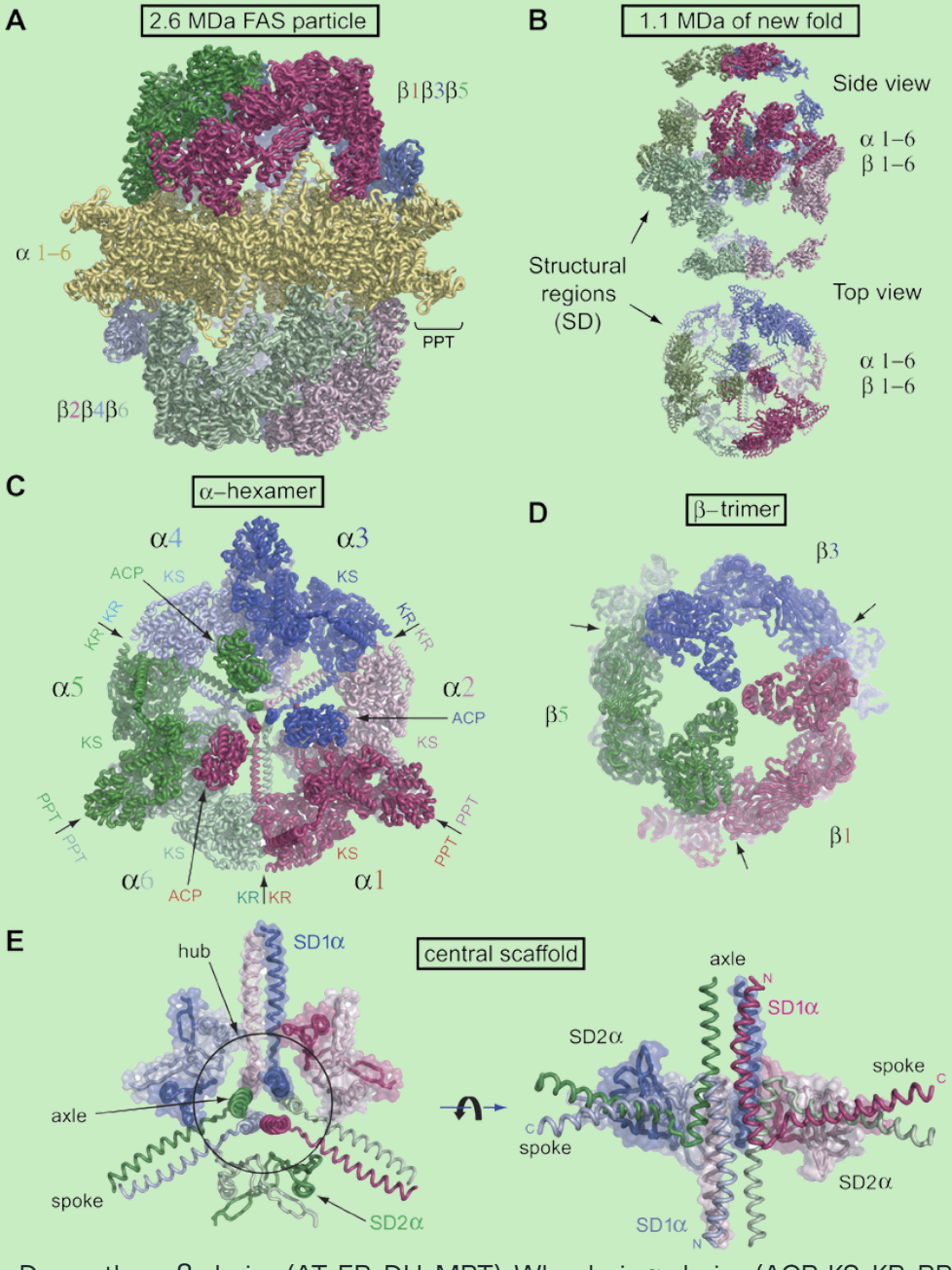
04

Why the Barrel Wins: two dimensions beat three

Being tethered means the carrier sweeps a surface instead of searching a volume. Lomakin and colleagues (2007) call this two-dimensional diffusion, and it is the main reason the barrel outperforms the loose-parts design. The cost is rigidity: a hard-wired barrel is hard to edit without disturbing its neighbours. That is part of why our edit has to be small.

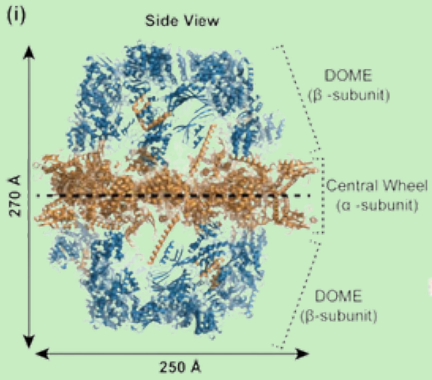
Quaternary Structure of the α6β6 Barrel

Twelve protein chains: six copies of an α chain, encoded by FAS2, about 210 kDa each, and six copies of a β chain from FAS1, about 230 kDa each. Together they make 2.6 MDa, 270 Å long and about 260 Å wide



The six α chains lock into a central wheel. The β chains form two trimers that dome over each face of the wheel like the caps of a drum. The wheel carries four of the working domains (ACP, KS, KR, PPT) and the domes carry the other four (AT, ER, DH, MPT). The two genes therefore split the labour, and our edit is on the α chain, in the wheel.

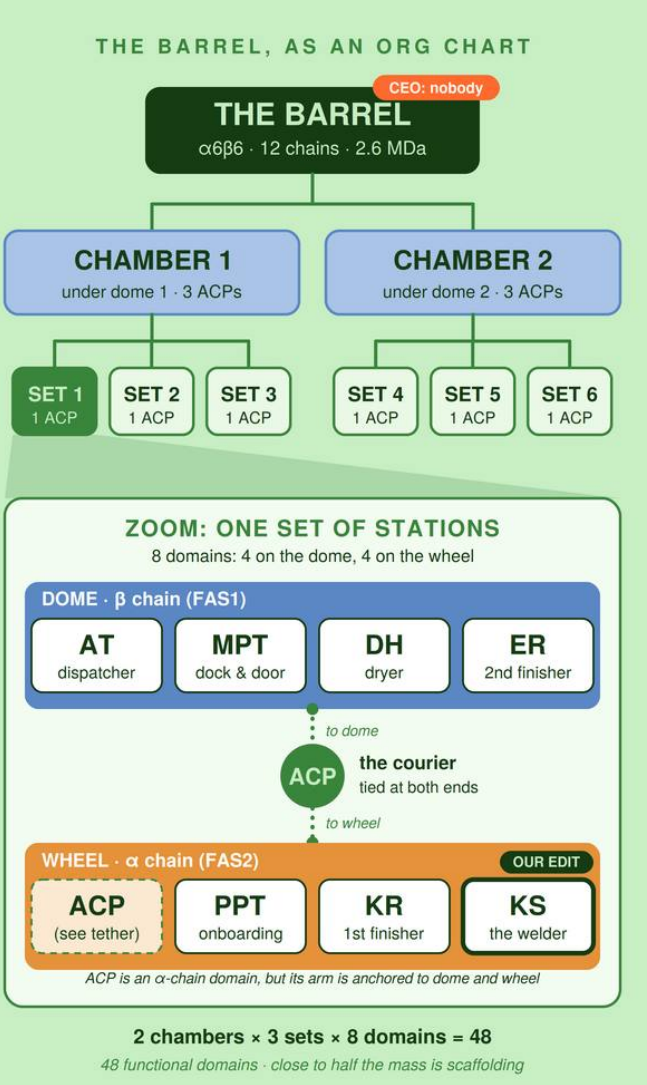
The complex has three-fold rotational symmetry, so each dome is a trimer and each chamber contains three identical reaction centres.



Dome: three β chains (AT, ER, DH, MPT). Wheel: six α chains (ACP, KS, KR, PPT).

Compartments and Reaction Centres

Twelve chains, two rooms, six sets of stations, six carriers.



Compartments: Two chambers, one under each dome. Each chamber holds three complete sets of stations and three copies of ACP, one per set. That makes six reaction centres and six ACPs in the whole barrel .

Tethering: Every ACP is anchored at both ends, one to the dome and one to the wheel, by a tether long enough to reach every station in its chamber. Six sets running in parallel means FAS is six small factories sharing one building, not one factory.

Catalytic vs Structural Mass: The Scaffold Fraction

FAS has 48 functional domains (6 sets × 8) and runs more than forty reaction steps. Close to half its mass catalyses nothing (Bukhari et al., 2014). That mass is scaffold, and its only job is to hold every working part in the right position relative to every other. Our project lives in the wheel. The domain we edit is KS, which forms the new carbon–carbon bond, and it sits on the α chain.

Eight domains. Four on the α chain, four on the β chain. Our project touches exactly one: KS.

Each reaction centre has the same eight domains: four on the α chain in the wheel and four on the β chain in the dome. Because the cycle behaves like a program, we label each domain with the instruction it executes. The analogy breaks where you would expect: there is no processor. No domain does anything until the carrier arrives. Seven of the eight never lengthen the chain; they load it, reduce it, dehydrate it, carry it, or (PPT) install the arm that carries it. Only one forms a new carbon–carbon bond, and that is the one we are editing.

The Catalytic Domains

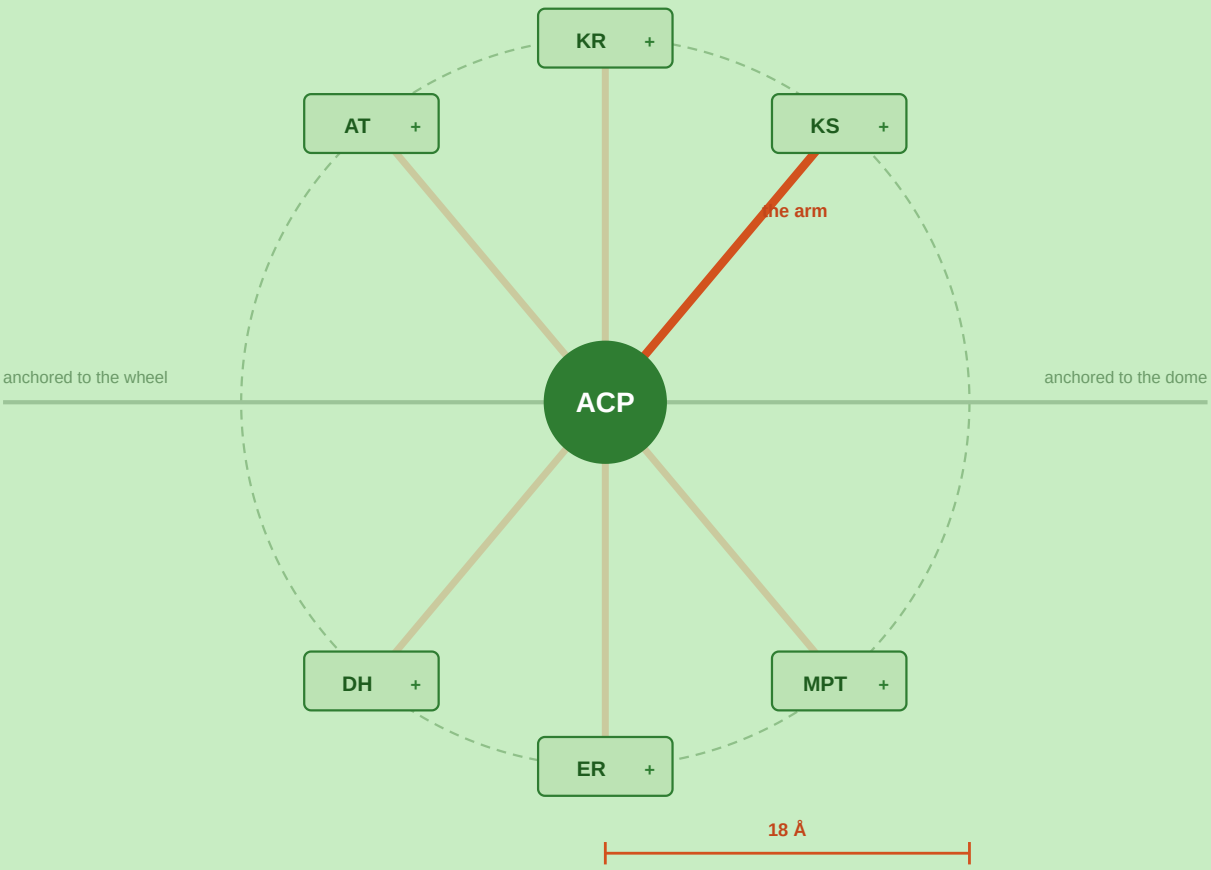
- AT, acetyltransferase (β) · LOAD_START. Transfers the two-carbon starter from CoA onto ACP.
- MPT, malonyl/palmitoyl transferase (β) · LOAD_EXTENDER and RELEASE. Loads every extender unit, and at the end transfers the finished chain off ACP.
- KS, ketosynthase (α) · CONDENSE. Joins the extender onto the growing chain by decarboxylative condensation. It makes the only C–C bond in the cycle.
- KR, ketoreductase (α) · REDUCE_1. Reduces the β-ketone to a β-hydroxyl using NADPH.
- DH, dehydratase (β) · DEHYDRATE. Removes water between the two reductions, leaving a double bond.
- ER, enoylreductase (β) · REDUCE_2. Reduces that double bond, using FMN and NADPH, leaving a saturated chain ready to repeat.
- ACP, acyl carrier protein (α) · the carrier. Holds the chain on its arm between stations and never lets go.
- PPT, phosphopantetheinyl transferase (α) · ACTIVATE. Attaches the arm to ACP in the first place (a post-translational modification).

The Phosphopantetheine Tether

The tether is the mechanical heart of the machine, and a feature synthetic biologists can reuse. It is a phosphopantetheine chain, 18 Å long, built from a piece of coenzyme A, flexible along its whole length and tipped with a sulfur atom. The growing fatty acid is bound to that sulfur for the whole synthesis and never floats free. Three principles make it work:

1. Bounded diffusion: a leash, not a search. ACP sits about 18 Å from the catalytic cysteine of KS, which is the length of its own arm. It is not told where to go; it simply cannot reach anywhere else.
2. Electrostatic steering: a lock that pulls the key. ACP's surface is negatively charged, and each station carries a positive patch beside its active site, the "electrostatic loading docks". ACP is not finding its way. It is being pulled.
3. Sequestration and release: the switchblade. Yeast ACP tucks part of the chain into its own hydrophobic core, and at the station the chain has to flip back out. This is the switchblade mechanism.

BOUNDED DIFFUSION · A LEASH, NOT A SEARCH



Everything the arm must reach lies inside the circle. Nothing else does — the arm is not told where to go, it simply cannot reach anywhere worse. Each station carries a positive patch (+).

Substrates and Stoichiometry

The Elongation Cycle

The Substrate Inventory: Fatty acid synthesis needs three inputs: acetyl-CoA, malonyl-CoA and NADPH. Acetyl-CoA is the two-carbon starter, used once per chain. Malonyl-CoA is the three-carbon extender. NADPH is the reducing power for the two reductions in every cycle. The barrel makes none of these itself. It only turns them into a chain. Think of the barrel as a machine that is handed an order form.

Input	Carbons	Where it comes from	Used per C16 chain
Acetyl-CoA	2	Central metabolism	1 (the starter)
Malonyl-CoA	3 (net 2)	ACC, outside the barrel	7
NADPH	n/a	Mostly the pentose phosphate pathway	14

Carbon Provenance and Energetics

Malonyl-CoA is made outside the barrel by a separate enzyme, ACC, which adds a CO_2 to acetyl-CoA at the cost of ATP. The first two carbons of every fatty acid come from acetyl-CoA, and every carbon after that comes from malonyl-CoA.

Spring-loaded bricks: Here is the neat bookkeeping. The CO_2 that ACC attaches is the same CO_2 that KS removes at the join. ATP is spent once, to "charge the spring" by carboxylating acetyl-CoA. The decarboxylation then releases that stored energy and pulls the carbon–carbon bond-forming step forward (a decarboxylative Claisen condensation). The three-carbon unit enters, one carbon leaves as CO_2, and the net gain is two.

The real gate: This is why ACC, not the barrel, is the true entry point to fat synthesis. Once malonyl-CoA exists, ATP has been committed. In yeast, ACC (Acc1p) is itself switched down by the energy-sensing kinase Snf1 when fuel is short, so the brakes begin one step before the barrel.

Net equation:
Acetyl-CoA + 7 malonyl-CoA + 14 NADPH + 14 H^+ → palmitate + 7 CO_2 + 8 CoA + 14 $NADP^+$ + 6 H_2O. In whole molecules: eight acetyl-CoA-derived units, seven ATP spent at ACC, fourteen NADPH.

"One acetyl-CoA, seven malonyl-CoA, fourteen NADPH. That is a fat"

Starter, Extender, Reductant

Three molecules enter every cycle. Two are building blocks and one is the electricity bill. For each: what it does, where it comes from, and what happens when it runs short.

01

Acetyl-CoA, the starter

Two carbons. Loaded onto ACP by AT, used exactly once per chain, and parked on the KS cysteine while the first extender is loaded. It is the first brick, so every fatty acid carries an acetyl-derived methyl end. Acetyl-CoA is also a metabolic hub, so fat synthesis competes with the TCA cycle and with protein acetylation for the same pool.

02

Malonyl-CoA, the extender

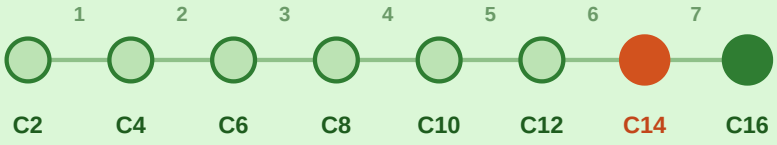
Three carbons, one of which leaves as CO_2 on joining. Made outside the barrel by ACC (Acc1p in yeast) and loaded onto ACP by MPT. It is the committed step into fatty acid synthesis: a cell that makes too little cannot lengthen a chain, however good its barrel is.

03

NADPH, the battery

Two per cycle, one for KR and one for ER. The cell makes most of its NADPH in the pentose phosphate pathway, so fat synthesis competes with other anabolic work for the same supply. When NADPH runs low, the γ subunit slows the barrel (Chapter 5), so the machine throttles itself instead of stalling mid-cycle.

SEVEN CYCLES TO SIXTEEN · SIX TO FOURTEEN



- C14 — our exit, myristic acid
- C16 — wild-type exit, palmitic acid

Each cycle skipped saves one malonyl-CoA and two NADPH.

Cycle Count and Chain-Length Arithmetic

The chain passes through C4, C6, C8, C10, C12, C14, C16. That is seven cycles, two NADPH each. Then MPT, the same enzyme that loaded the three-carbon units, transfers the finished chain from ACP to free CoA.

Sixteen carbons takes seven cycles. To get fourteen, the chain must leave after six. In material terms, a C14 chain consumes one acetyl-CoA, six malonyl-CoA and twelve NADPH, so a shorter chain is also a cheaper one. Each cycle skipped saves one malonyl-CoA and two NADPH. Whether that saving matters to the cell, we do not yet know.

Steps 1–5: Priming and Condensation

Loading the starter. The acetyltransferase (AT) domain specifically recognizes acetyl-CoA and transfers its two-carbon acetyl group directly onto the terminal sulfhydryl group of the acyl carrier protein's (ACP) flexible phosphopantetheine arm. Once loaded, the initial two-carbon unit is positioned securely within the catalytic barrel assembly to initiate the primary synthetic cascade.

Parking the starter. The newly loaded ACP arm swings toward the ketoacyl synthase (KS) domain. Through a transacylase reaction, KS accepts the acetyl group onto its critical active-site cysteine residue, freeing the ACP arm to retrieve incoming extension units.

Making the extender. Outside the core enzyme barrel, cytosolic acetyl-CoA carboxylase (ACC) synthesizes malonyl-CoA by carboxylating acetyl-CoA using dissolved carbon dioxide and ATP. This newly formed three-carbon malonyl extender unit freely diffuses into the local active site region.

Loading the extender. The malonyl/acetyl-transferase (MPT) domain selects a cytosolic malonyl-CoA molecule and transfers its malonyl moiety onto the now-empty ACP sulfhydryl arm. Crucially, this versatile MPT domain is also responsible for executing the final chain-unloading step once synthesis reaches completion.

Condensation. ACP swings back to the active KS domain, where the parked acetyl group undergoes a decarboxylative condensation with the malonyl group. The carboxyl group escapes as a carbon dioxide molecule, generating a four-carbon β-ketobutyryl chain attached to ACP; this pivotal step represents the sole mechanism driving carbon chain elongation.

Steps 6–10: Reduction and Iteration

First reduction. The ACP arm swings its attached four-carbon β-ketobutyryl intermediate to the ketoacyl reductase (KR) domain. Using NADPH as an electron donor, KR reduces the β-keto group into a hydroxyl group, yielding a β-hydroxybutyryl moiety.

Dehydration. The intermediate translocates to the dehydratase (DH) domain, which eliminates a molecule of water across the carbon-carbon bond. This chemical dehydration introduces a trans double bond, converting the intermediate into a four-carbon unsaturated butenoyl-ACP complex.

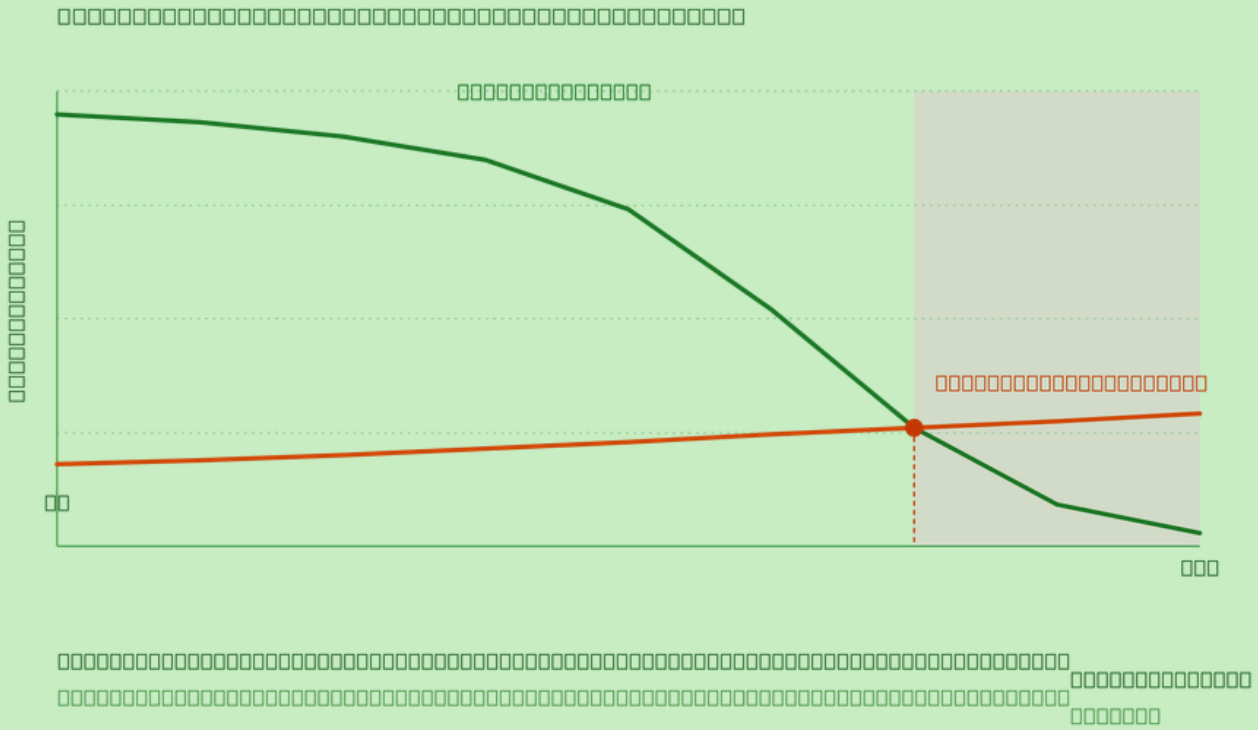
Second reduction. The enoyl reductase (ER) domain, utilizing its bound FMN cofactor and a second molecule of NADPH, reduces the double bond to produce a fully saturated four-carbon butyryl chain, concluding the initial reduction cycle.

Second cycle. The fully saturated butyryl group is transferred back to the KS active-site cysteine, MPT loads a fresh malonyl-CoA extender onto the ACP arm, and KS catalyzes another condensation to form a six-carbon intermediate.

Iterate to termination. This iterative cycle repeats continuously, adding two carbons per turn, until a sixteen-carbon palmitoyl chain is assembled, whereupon MPT unloads the product onto CoA; premature release at fourteen carbons requires MPT offloading to outcompete KS condensation during the seventh cycle.

CHAIN-LENGTH CONTROL 2026

Why sixteen? There is no timer and no stop signal.



Rejecting the Molecular Ruler

The KS domain is often said to contain a "molecular ruler". It is a useful phrase and wrong in a useful way. A ruler measures a length, reports a number, and something acts on the number. Nothing here works like that. There is no counter and no step that checks "sixteen reached".

What exists is a race. The chain has to thread into a narrow tunnel to reach the site where it is extended, and every extra two carbons makes the threading a little harder. Meanwhile release is always possible. Elongation and release are competing reactions, and near sixteen carbons the competition flips. Sixteen is not a rule the enzyme follows; it is where the machine runs out of odds.

Picture a loop with no counter, in which every pass tosses a coin: continue or exit. The coin is fair for short chains and becomes more heavily loaded toward exit as the chain grows. If the odds stayed constant, product lengths would spread out in a long tail. Real FAS gives a sharp peak at C16–C18, so the odds of continuing must collapse near that length.

The Gatekeeper: Met1251 as a Conformational Gate

At the narrow midpoint of the substrate-binding tunnel sits a single conserved residue, Met1251, which functions as the primary conformational gatekeeper. Rather than obstructing the channel outright, this residue exerts dynamic control over acyl chain entry. As an incoming acyl chain enters the binding pocket, interaction with the Met1251 side chain induces a precise rotational displacement, opening the inner hydrophobic cavity to accommodate further chain extension.

Conceptually, this mechanism functions like a molecular turnstile: the acyl chain does not forcibly push past the barrier, but rather energetically persuades the flexible side chain to rotate. As the chain length increases beyond optimal steric limits, this persuasive interaction becomes progressively less favorable. Consequently, longer acyl chains exhibit significantly reduced efficiency in triggering the conformational opening required for continued elongation.

The Opposing Wall: Phe1279

The architecture of the binding tunnel is defined by a dual-sided constraint. Directly opposite Met1251, the aromatic side chain of Phe1279 projects into the channel lumen from the opposing wall (Gajewski et al., 2017; Tasaki et al., 2026), creating a tight spatial bottleneck. Because both sides contribute to spatial constriction, the passage can be narrowed through structural modifications on either wall.

This arrangement offers two distinct mechanical strategies for restricting turnstile movement: either directly inhibiting the rotational freedom of Met1251, or physically shrinking the volume of the space into which it must rotate. Implementing either structural strategy increases steric hindrance, systematically shifting product specificity toward shorter-chain fatty acids such as myristic acid (C14).

Engineering Levers: Two Ways to Restrict the Tunnel

To rationally engineer chain-length specificity, two distinct structural levers can be applied across the binding channel.

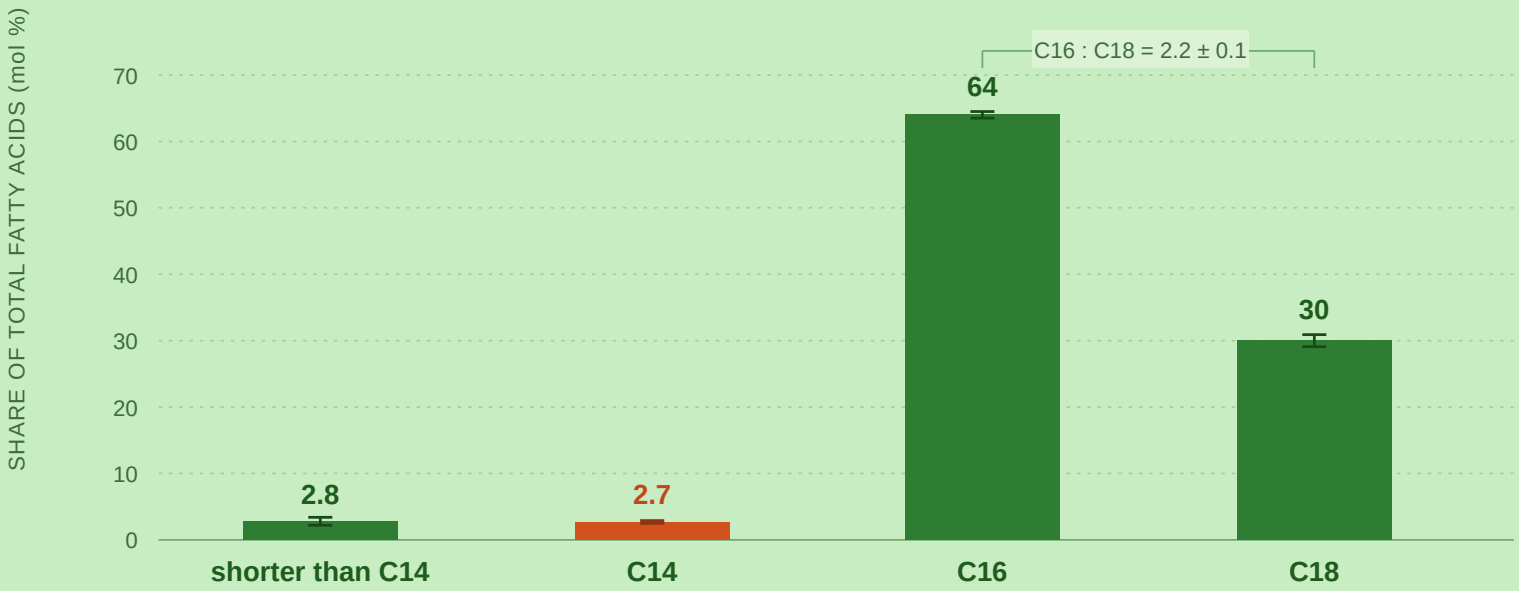
- Lever 1 (Restrict the gate): Introducing steric bulk adjacent to the gatekeeper residue via G1250S and G1253A substitutions physically constrains Met1251, restricting its rotational freedom and preventing the gate from opening fully.
- Lever 2 (Narrow the far wall): Bulky substitutions on the opposite face, such as F1279Y and I1280F, reduce total cavity volume and eliminate the spatial clearance needed for side-chain rotation.

Together, these modifications constitute the primary molecular toolkit for tunnel engineering. Our current efforts focus specifically on optimizing the first lever.

Where the Gate Closes in Wild Type

In wild-type *Saccharomyces cerevisiae*, acyl chain elongation does not terminate at fourteen carbons. Independent investigations across multiple laboratory strains demonstrate consistent chain-length distributions (Oura & Kajiwara, 2006; Sangwallek et al., 2013). Under physiological conditions, the conformational gate effectively restricts release until the chain reaches 16 to 18 carbons.

Consequently, myristic acid (C14) accounts for less than 3 percent of total cellular fatty acids, whereas C16 and C18 species predominate, comprising 94–97 percent of the final pool. A detailed quantitative breakdown of these wild-type profile proportions is provided on the following page.



Wild-type product spectrum. *S. cerevisiae* W303-1A. Mean ± s.d., mol % of total fatty acids. Data from Sangwallek et al., 2013. C14 is the fraction our V1254W strain has to move.

Reading the Gate in Yeast: the Wild-Type Product Spectrum

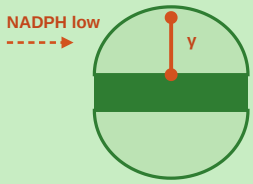
In the wild type, C14 sits at 2.7 ± 0.2 mol percent, about equal to everything shorter than it combined. That baseline sets the bar for our own result. A successful V1254W strain must raise the C14 fraction clearly above this background, measured by GC-MS in the same strain background, under the same growth conditions, and against an empty-vector control. We will report it as a shift in the C14 fraction, not as an absolute titre.

Ok, let's take a break. Why was the fatty acid so calm?
Because it had reached a state of palmitation.

Regulation of FAS: Three Layers of Control

LENS: CONTROL ENGINEERING. THREE BRAKES ON ONE MACHINE, AND NO TWO WORK THE SAME WAY.

01

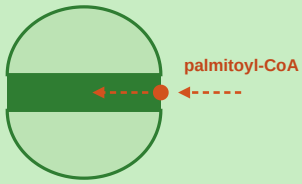


The γ subunit

LOAD-SENSING GOVERNOR

Senses NADPH. Halves initial turnover and suppresses futile cycling at ER.

02



The product site

FEEDBACK SENSOR

Binds palmitoyl-CoA on the outer wall. Shuts down KR inside.

03



Phosphorylation

REVERSIBLE FIRMWARE

Tracks the cell's life cycle. Shifts output toward C18.

Why a 122-ATP Machine Cannot Run Open-Loop

A machine that costs 122 ATP per unit does not run unsupervised. FAS has at least three control layers. We use control-engineering names for them: a governor, a feedback sensor and firmware. Fun fact: one of the three went unnoticed for about fifty years.

The γ Subunit: A Load-Sensing Governor In 2020 a group in Göttingen purified yeast FAS more gently than usual and found a 20 kDa protein riding along that nobody had catalogued: Tma17p, renamed the γ subunit. It had always been in FAS preparations, held on weakly enough that ordinary salt washes stripped it off before anyone looked. Six copies bind one FAS, one per reaction centre, each touching one α chain and two β chains. It does not switch FAS off. When NADPH runs low it halves the initial turnover rate and suppresses futile cycling at the ER station, like a rev limiter that engages when fuel runs short. Its binding also rotates the whole complex, so ACP parks at AT instead of KS.

The Product Sensor on the Outer Wall

In late 2025 a group in Toronto found something stranger. The finished product, palmitoyl-CoA, binds a site on the central wheel, on the outside of the barrel and nowhere near any active site, and from there shuts down the ketoreductase step inside (Hasan et al., 2025). The signal travels through ordered water: binding changes how water is arranged around the site, and that change propagates to the catalytic machinery. Think of a pressure sensor on the outside of a tank that closes a valve on the inside through a hydraulic line made of water. The site is conserved across several pathogenic fungi, which makes it a drug target.

Phosphorylation: Reversible Firmware

A 2020 study tracked 10,794 phosphorylation sites across the yeast life cycle. FAS was among them, and phosphorylation changed the length of the fat the cell makes (Martínez-Montañés et al., 2020). Locking three sites on FAS2 into the phosphorylated state shifted output toward C18.

The cell already has a reversible dial for chain length and uses it, retuning its fat across its life without touching its DNA. That is a firmware update. Our edit is a hardware change: a permanent mutation doing the same job. The firmware will still be running afterwards, and we did not install it.

BOTTOM RIGHT, The Three Levers (small table):

Lever	Mutation	Domain	Logic
Change the intake	I306A	AT	Widens the starter-block channel
Change the gate	G1250S, M1251W, F1279Y	KS	Obstructs long-chain elongation
Change the exit	R1834K	MPT	Weakens malonyl binding, so release wins sooner

Prior Engineering: People Who Have Broken This Before

The gate was located by swapping parts before anyone mutated a residue. Replace the α chain of *S. cerevisiae* with the one from Hansenula polymorpha, which prefers C18, and the C16/C18 ratio drops from 2.2 to 0.5. Narrowing the swap down points to the KS domain as the main carrier of the effect .

In 2017, Gajewski and colleagues pulled three levers (as described in Wang et al., 2026), and together they gave about a 23-fold increase in short and medium-chain output.

The Three Levers

To rationally engineer fatty acid synthase specificity across the pathway, three distinct mechanistic levers can be strategically targeted throughout the enzyme complex. The first lever focuses on changing the intake; introducing the I306A substitution within the acetyltransferase (AT) domain successfully widens the starter-block channel to alter substrate access. The second lever aims to change the gate mechanism within the ketoacyl synthase (KS) domain. By implementing specific point mutations—namely G1250S, M1251W, or F1279Y—steric bulk is strategically added to directly obstruct long-chain acyl elongation beyond desired lengths. Finally, the third lever modifies the exit dynamics via the malonyl/acetyl-transferase (MPT) domain. Applying the R1834K mutation deliberately weakens malonyl binding affinity, which accelerates offloading kinetics so that product release successfully outcompetes continued elongation sooner in the iterative cycle.

Elongation and release are racing. Slow the runner, move the finish line, or trip the runner. All three work.

Our Design: Engineering C14

Computation turns electricity into heat. Removing that heat is now a materials problem, and every coolant has a flaw.

Mineral oil is cheap, viscous and flammable. PFAS fluids perform well, cost hundreds of dollars a gallon and are being phased out. Plant oils biodegrade, then oxidise into sludge, because the double bonds that keep them liquid are the same bonds that let them go rancid. (The weak spot is the C–H bond next to a double bond, which is easily attacked by oxygen.)

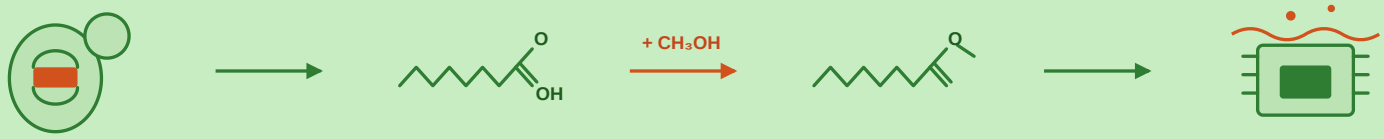
Methyl myristate avoids all three problems. It is the methyl ester of myristic acid: fourteen carbons, fully saturated, so there is nothing to oxidise. It is a dielectric, so it does not conduct. It is thin enough to pump. Saturation also raises the melting point, and methyl myristate melts near room temperature (about 19 °C), so formulation and blending will be part of the engineering. We state that openly.

Nearly everyone in Chapter 5 was chasing C6 to C10. We want C14, which sits between where the gate normally closes and where these mutants push it. Almost nothing makes C14 as its main product.

C14 Is Not a Fantasy

Milk fat is naturally rich in short and medium-chain fatty acids, up to exactly C14. Butter is the most widely eaten proof that a C14-containing fat is stable and attainable.

Fun fact: myristic acid is named after nutmeg, Myristica fragrans.



Yeast
FAS2 V1254W

Myristic acid
C14:0, made by the cell

Methyl myristate
the methyl ester

Immersion coolant
dielectric, non-oxidising

IN THE YEAST

DOWNSTREAM CHEMISTRY

The ester step is a downstream chemical conversion, not performed by the cell.

Convergent Evidence: Three Case Studies

Case 1 · THE OG Biologists, Brewers: Cerulenin-Resistant Sake Yeast

Ginjo sake smells of apple. That aroma is ethyl caproate, with ethyl caprylate beside it. Both are esters of medium-chain fatty acids (C6 and C8), which is what you get when yeast FAS stops early. Brewers have selected for this by taste for generations, with no idea what a ketosynthase was. Cerulenin inhibits FAS, and yeast that survive it usually carry a mutation in the acyl-binding tunnel. Cerulenin works by covalently modifying the catalytic cysteine of KS, so resistance selects for tunnel variants.

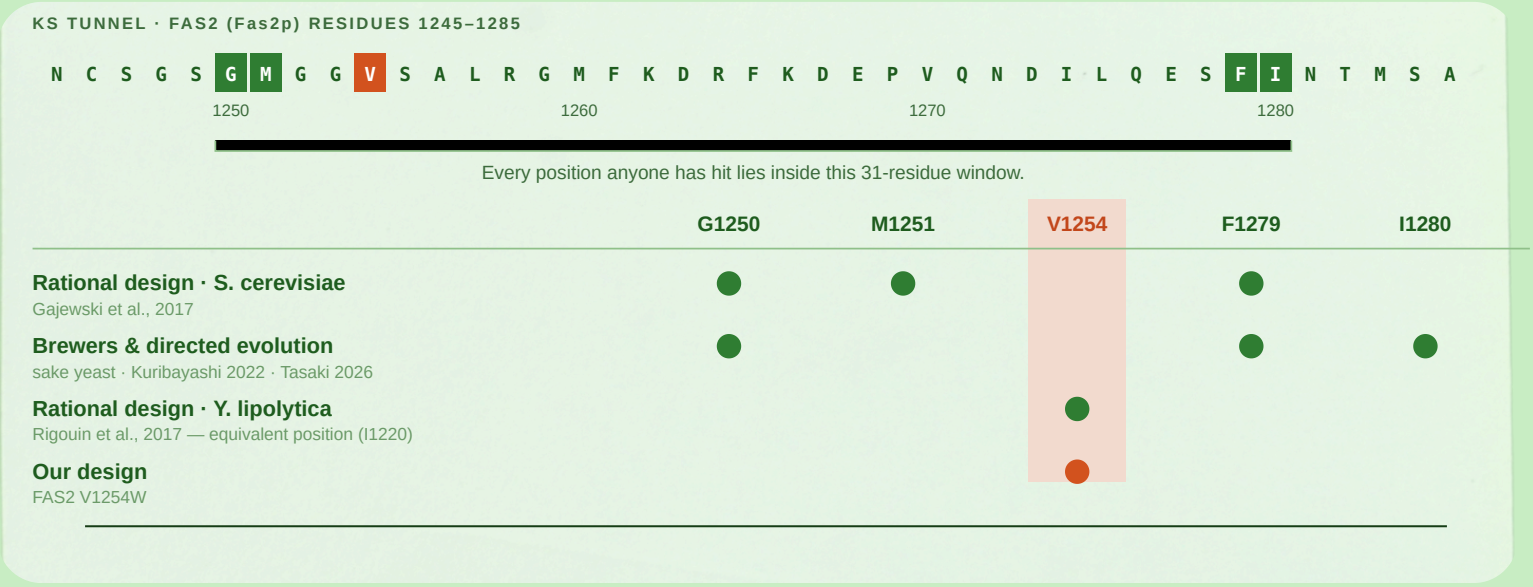
In 2026 a cerulenin-resistant sake strain yielded FAS2-1280F, a bulkier residue at Ile1280 beside Phe1279. It produced 1.9 times more caprylic acid and put ethyl caprylate into a sake that previously had none (Tasaki et al., 2026). The same few residues, all between 1250 and 1280, keep turning up: found by structural biology, by directed evolution and by brewers.

Case 2 · Bringing a blade to a sumo wrestling fight

The newest idea ignores the tunnel. Each dome has room for an extra protein of up to about 150 kDa, so the Göttingen group fused a bacterial thioesterase onto the γ subunit and parked it inside the reaction chamber (Singh et al., 2023). The result was more C8 and C10 and less C12, C14 and C16. They never touched the gate; they put a blade in the room.

Case 3 · Smile, you're in an earthquake: The Structural Biology of FAS

FAS is fragile and never holds still, and photographing it took decades of getting gentler. The first yeast crystal structure, in 2007, came from a lab that was after a ribosome and ended up with the wrong protein. Later, cryo-electron microscopy kept losing the most delicate piece, PPT, because the air–water interface of the thin ice film damaged the protein. One fix is to support the sample on a sheet of graphene one atom thick. By 2023 the maps were sharp enough to place 7,700 water molecules. What else in a cell is too delicate to photograph, and what would you put under it?



Design Rationale: FAS2 V1254W

Hypothesis. V1254W replaces valine 1254 with tryptophan in the KS domain of Fas2p. Position 1254 is on the Met1251 side of the tunnel, next to G1253 and three positions from the gatekeeper itself. In the turnstile model, our edit should belong to Lever 1, restricting how freely Met1251 can rotate, like G1250S and G1253A. Prediction: a measurable shift of output from C16/C18 toward C14. GC-MS will tell us whether the hypothesis holds.
Rationale. The equivalent position in *Yarrowia lipolytica* was changed to a bulky aromatic residue and C14 output rose. V1254W is that substitution in *S. cerevisiae* numbering (Rigouin et al., 2017).

Tuning, Not Breaking

Why a moderate substitution: a dimmer, not a breaker. This looks conservative, and it is deliberate. The aggressive mutants, such as M1251W and its combinations, push output down to C6 and C8, and a yeast that cannot make enough C16 and C18 must be fed fatty acids to survive. C16 and C18 are what membranes are made of, and a cell that cannot make them cannot build itself. We are moving the exit by two carbons, not ten: retuning the machine, not demolishing the part that makes the walls.

Construction

In silico. We took the 1.9 Å structure of S. cerevisiae FAS and modelled C14, C16 and C22 into the KS tunnel to compare how the tunnel accommodates chains of different length.
In the lab. We built the mutant gene and cloned it into p426GPD, a small circular piece of DNA (plasmid) that yeast can carry. The gene was cut and pasted in with restriction enzymes in two rounds, with a gel check after each. We amplified the plasmid in E. coli DH5α, transformed it into *S. cerevisiae* BY4742, and selected plasmid-carrying cells on SD-URA, a medium on which only they can grow.

Thank You

It is with immense joy, pride, and a touch of deep emotion that we release this booklet.
We extend our deepest gratitude to our faculty advisors, Prof. Rajesh Patkar and Prof. Saket Choudhary, whose constant guidance, wisdom, and encouragement navigated us through every step of this project. We are equally grateful to our mentors, host laboratories, and generous sponsors, whose invaluable resources, technical insights, and unwavering support made this work possible.
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Finally, this milestone would be incomplete without recognizing our incredible team. To our seniors, who built the foundation and set the standard, and to our juniors, whose passion and energy breathe life into every project—thank you for your dedication, late nights, and belief in this vision.
While this booklet is a collective labor of love, any errors or oversights contained within remain entirely our own.

Date:
21/10/2026

Courtesy of:

Durva Saraf
Wet Lab Head 2026

Ranit S. Sooraj
Sr. Wet Lab Researcher

Aayush Gupta
Jr. WetLab Researcher

Angel Singhvi
Team Lead 2026

Aarohi Gupta
Team Lead 2026

Glossary

Key terms of fatty acid synthase, grouped by role.

01 BUILDING BLOCKS

Acetyl-CoA
The two-carbon starter unit that begins every chain.

Malonyl-CoA
The three-carbon extender, used every cycle. It loses one carbon as CO_2 on joining, so the net gain is two.

NADPH
The cell's main reducing-power carrier. FAS spends two per cycle.

02 CARRIER MACHINERY

ACP
Acyl carrier protein
The tethered carrier that moves the growing chain between stations.

Ppant arm
Phosphopantetheine
The 18 Å flexible phosphopantetheine tether, tipped with a sulfur, that ACP uses to hold the chain.

03 ARCHITECTURE

AT, MPT, KS, KR, DH, ER, PPT
The seven enzymatic domains, described on page 11.

$\alpha_6\beta_6$
Barrel stoichiometry
The barrel's stoichiometry: six α chains and six β chains.

Scaffold
Structural protein that catalyses nothing and only holds the catalytic domains in position. About half the barrel.

04 REGULATION & CONTROL

Gatekeeper residue
The amino acid in the KS tunnel whose size and mobility help set how long a chain grows before release. Met1251 in yeast.

Allosteric
Acting at a distance: binding at one site changes activity at another site in the same protein.

γ subunit
Tma17p
The regulatory protein, one per reaction centre, that slows FAS when NADPH runs low.

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